CS5044: Practical 2 (group work)

This practical is to be completed in groups of three students. The goal of this practical is to *collaboratively* design and implement a web-based visualisation of a data set of your choice using D3.js. Your visualisation should reflect on what you have learnt about designing expressive and effective visualisations, also considering interactivity (see, for example, Shneiderman's visualisation mantra [1]).

WEIGHTING AND MARK DESCRIPTORS

This practical is weighted at 50% of the coursework component (and therefore **30% of the overall grade**). **Deadline April 23, 2026, 21:00 UK time.** The data set, sketches, final visualisation, and the report should be submitted to MMS. MMS is the definitive source for deadline and credit details.

RULES

You are expected to work on the practical together in groups of three, following the Good Academic Practice policy:

https://www.st-andrews.ac.uk/education/handbook/good-academic-practice/

Lateness penalties follow the default School policy (Scheme A: 1 mark per 24-hour period, or part thereof). You can review the policy on lateness penalties here:

https://info.cs.st-andrews.ac.uk/student-handbook/learning-teaching/assessment.html#lateness-penalties

PRACTICAL DESCRIPTION

In this practical, you will *collaboratively* develop a concept for a web-based visualisation and implement it using D3.js – https://d3js.org/. The requirements for this practical are described below.

GROUP MEMBERSHIP

This assignment is to be done in groups which will be assigned by the module lecturers. If you have any serious reasons as to why you should NOT be in the same group as someone else, please contact a Student Welfare Officer as soon as possible, and in any event, within a week from the date of releasing this practical, by which time we would like to have groups formed. If you are unable or unwilling to raise a concern before groups are allocated, you can still contact a Student Welfare Officer within one week of the groups having been announced. Note that in this case, it may be much harder to make group changes.

WEB-BASED VISUALISATION [collaborative]

You should come up with a design and implementation of a web-based visualisation based on a data set of your choice (the requirements on the data are specified below). Your visualisation design should be driven by particular questions that you would like your visualisation to help answer and related visual analysis tasks that you would like it to support (see Munzner, Chpt. 3 [2]). It is up to you to specify these questions and tasks. **Your visualisation should answer at least one open-ended question that you specify in your report.**

Your visualisation design should combine and balance clarity and creativity, making the key aspects of the data both visible and easy to explore. Your visualisation will be assessed based on the following.

- How well does it suit your specified question(s) and task(s)?
- Your choice of visual encoding, using visual variables in an effective and expressive way (see Munzner, Chpt. 5 [2]).
- How you support basic interactivity and tasks as specified by Shneiderman [1]. Note that not all these low-level tasks have to be supported – this depends on your data set and questions. However, your visualisation should at least provide an "overview" and some "details on demand" – whatever this may mean for your particular data and visualisation.

It is completely fine to base your visualisation design on existing visualisation techniques that you know, but make sure to name and reference these in your report.
Your visualisation concept should be the result of an ideation process in the form of paper and/or digital sketches. Your sketches should demonstrate that you have thought of multiple alternative visualisation ideas. Your final visualisation should be presented in a way that clearly describes the visual mapping of data, interactive features, and implementation considerations.

IMPLEMENTATION GUIDELINES

You should implement your visualisation using D3.js. You may use other JavaScript libraries alongside D3.js where appropriate (e.g., jQuery, Leaflet, Crossfilter, etc.). However, you must **NOT** use libraries that are built on top of D3.js and effectively generate charts for you (e.g., DC.js), nor higher-level, declarative charting/visualisation libraries that abstract away the visual encoding and interaction design (e.g., Plotly.js). If you are unsure whether a library is acceptable, please ask the lecturers in advance.

Your implementation should be developed using HTML, CSS, and JavaScript (vanilla JS), following best practices for web-based visualisation. You must **NOT** use component/SPA frameworks like React.js or Vue.js, as the focus of this practical is on direct interaction with D3.js.

A key aspect of this practical is that you program the visualisation yourself - the use of Tableau or Excel is ok for sketching, but not for the final visualisation. You may use and modify code that you find online or in books. However, make sure to explicitly acknowledge and reference all sources that you have used to implement your visualisation, **both on the project webpage and in the form of code comments**.

Implementing a web-based visualisation from scratch is not a trivial task. A key aspect of this practical is to develop a design suitable for your question and data, yet easy enough to be implemented in the provided time. It may be a good idea to come up with a simple design first and then iteratively refine this design, adding more complex features if there is time (see Process section further below).

Make sure to include an appropriate title and a brief introductory paragraph on the webpage that introduces the visualisation and its aims. Also, make sure to include appropriate labels and legends. Your visualisation should be usable by someone who has not read your practical report!

REPORT [collaborative]

Your report should include key information about the underlying data and how to read and interact with the visualisation. The report should state the word count, which must not exceed 1000 words. Failure to adhere to the limit in the report is penalised according to Scheme A of the University policy on coursework penalties (1 mark for work that is 10% over-length, then a further 1 mark per additional 10% over).

> https://www.st-andrews.ac.uk/policy/academic-policies-assessment-examination-and-award-coursework-penalties/coursework-penalties.pdf

Make sure to provide images and/or screenshots of your visualisation to illustrate your argument. The following sections should be included.

DATA & QUESTIONS

Briefly describe the data set that your visualisation is based on, including key attributes and attribute types, number of data points, etc. Describe the question(s) and visual analysis tasks that you would like to support with your visualisation. Provide links to the original data sources (if applicable).

DESCRIPTION OF VISUALISATION

Briefly describe the design of your visualisation and explain how it should be read. Specify which visual variables have been used to encode the data attributes and provide a clear description of how users can read and interact with your visualisation. Describe and illustrate all interactive features. Explain how the interactive elements of your visualisation function and what impact they have on the user experience. Reference existing visualisation techniques that relate to your visualisation design. This section should serve as a 'how-to' guide for reading and interacting with your visualisation.

IMPLEMENTATION

Briefly comment on how you have implemented your visualisation, including links and references to any outside sources that you have used to implement your visualisation. This could be provided in list form. Details about the implementation could also be merged with your description of the visualisation concept; you do not necessarily have to provide a separate section.

INSIGHTS FROM THE VISUALISATION & CRITICAL DISCUSSION

Briefly describe the insights people can gather from your visualisation. Make sure to not only answer and illustrate the questions that can be answered, but also briefly describe the exploration steps necessary to arrive at these insights. Also, discuss the limitations of the visualisation (both regarding its design and implementation) and how these could be overcome.

EVIDENCE OF IDEATION PROCESS [collaborative]

In addition to the project webpage and your report, you will submit evidence of your ideation process in the form of a PDF file. Include the different visual representations and designs you came up with before settling on a particular concept. This part of the report could consist mostly of sketches with brief comments, but it should be clear how your process led to the final visualisation concept.

We recommend using a structured ideation framework to guide this process. One well-established approach in visualisation design is the Five Design-Sheets (FdS) method, which supports iterative development. However, whatever approach you choose must provide clear evidence of the core components that FdS is designed to elicit, especially an explicit stage of divergent thinking that produces multiple distinct design alternatives, followed by a rationale for selecting and refining your final concept.

PERSONAL STATEMENT [individual]

Each group member should submit a personal statement describing their role in this project, a brief reflection on the visualisation process and the final results, and what they have learnt as part of this practical. The personal statement must not exceed 250 words. Failure to adhere to the limit in the personal statement is penalised according to Scheme A of the University policy on coursework penalties (1 mark for work that is 10% over-length, then a further 1 mark per additional 10% over).

POSSIBLE EXTENSIONS

The following are optional examples of extensions that may strengthen a submission; students are not required to implement them, may propose their own appropriate extensions, and extensions are not necessary to achieve a top grade if the core requirements are met to an excellent standard.

- A video of the visualisation clearly and concisely describes and demonstrates its design, interactive features, and insights that the visualisation provides. The video should not exceed 3 minutes or the MMS's 600 MB upload limit!
- Advanced interactive features such as coordinated views [3], brushing and linking, etc…

NOTES ON DATA & POTENTIAL DATA SOURCES

Data Requirements and Selection

Your data should include at least **100 data points** and **four different attributes** (e.g., "Country", "latitude", and "longitude" would count as a single attribute focusing on "location"). **If in doubt, talk to the lecturers about the data that you would like to work with.**

Based on the data, you should come up with a question that your visualisation should enable the user to explore. The question could be quite focused or more open-ended. Make sure to also consider who is going to interact with the visualisation and what type of goals the user may want to achieve.

Potential data sources are provided below, but feel free to find other sources. Follow your interest! **When you find data that you would like to use, make sure you have the right to use it! Look for copyright constraints, etc.**

Datasets

DATA IS PLURAL
A weekly newsletter of useful/curious datasets.

The letter archive: Data Is Plural (data-is-plural.com)

Spreadsheet with all archived posts:
https://docs.google.com/spreadsheets/d/1wZhPLMCHKJvwOkP4juclhjFgqIY8fQFMemwKL2c64vk

OECD.org
Data sources by the Organization for Economic Co-operation and Development (OECD)
https://data.oecd.org/

UNESCO Institute for Statistics
http://uis.unesco.org/en/home

DATA.GOV.UK
Data published by government departments and agencies, public bodies and local authorities.
https://data.gov.uk/

EU Open Data Portal
http://data.europa.eu/euodp/en/data/

NHS Health and Social Care Information Centre
http://content.digital.nhs.uk/home

FiveThirtyEight
Opinion poll analysis, politics, economics, and sports blogging.
http://fivethirtyeight.com/

Kaggle Data Sets
https://www.kaggle.com/datasets

OFCOM digital inclusion research and data
https://www.ofcom.org.uk/phones-and-broadband/accessibility/exclusion/
https://www.ofcom.org.uk/research-statistics-and-data/

GOOD THINGS FOUNDATION digital inclusion datasets
https://www.goodthingsfoundation.org/policy-and-research/research-and-evidence/research-2024/digital-inclusion-datasets

Datasets on Child Poverty and Deprivation

For this year's project, we encourage you to explore child poverty and deprivation as a theme for your data visualisation. This is a suggested area of focus, not a requirement. Choosing this topic is optional and will not affect grading.

If you're exploring child poverty, the UNICEF State of the World's Children (SOWC) 2025 statistical tables provide a rich set of country-level indicators that relate to material deprivation, education (e.g., learning poverty), nutrition, social protection, and related outcomes. However, you are not limited to this source; you may use any dataset that meets the data requirements. More about the suggested theme can be found in Appendix 1 at the end of this document.

UNICEF SOWC 2025 statistical tables
https://data.unicef.org/resources/sowc-2025/

DELIVERABLES

The deliverables of this practical are listed below. Submit a single .zip file to MMS containing the following:

- All source files for your visualisation and underlying webpage (HTML, CSS, JavaScript, and any required libraries).
- A README file explaining how to run your visualisation.
- Your report as a PDF.
- Your personal statement as a PDF.
- Your sketches/ideation process documentation as a PDF (with brief comments where necessary).

You must ensure that your code can be compiled and run on the School's lab machines before submitting. Please refer to the general coursework requirements for further details:

https://info.cs.st-andrews.ac.uk/student-handbook/learning-teaching/assessment.html#Coursework

ASSESSMENT

Your marks for this practical will be determined by the quality of your web-based visualisation, project webpage, report (which should include justifications of your design and critical reflections on your visualisation outcomes), and personal reflection on the project. The School's general mark descriptors apply and can be reviewed here:

https://info.cs.st-andrews.ac.uk/student-handbook/learning-teaching/feedback.html

Indicative guidance is provided below on examples of what is expected within each mark range.

A ***practical in the 11-13 range*** will include a basic web-based visualisation providing an overview of the selected data, plus detail views to provide basic insights into the given question. The report will provide a good overview of the project, including a reflection on the strengths and limitations of the visualisation.

A ***practical in the 14-16 range*** will include a well-designed interactive web-based visualisation that provides a good overview of the data set from several perspectives, including details-on-demand. Both visual encoding and interaction techniques will be well thought through. The visualisation will be embedded in a webpage that provides additional information to help interpret and interact with the visualisation. A good practical will include a well-written and structured report that includes a clear description and justification of design decisions visible in the visualisation and a critical discussion of the strengths and weaknesses of the visualisation.

A ***practical in the 17+ range*** will feature all of the above, with an excellent visualisation concept, design and implementation, including well-designed advanced interactive features, and/or one or multiple extensions (e.g., additional views, video…).

RESOURCES

[1] Ben Shneiderman. The Eyes Have It: A Task by Data Type Taxonomy for Information Visualisation. In Proceedings of the IEEE Symposium on Visual Languages, 1996 https://www.cs.umd.edu/~ben/papers/Shneiderman1996eyes.pdf
[2] Tamara Munzner. Visualisation Analysis and Design, Chapter 3: Why? Task Abstraction, 2015
[3] Tamara Munzner. Visualisation Analysis and Design, Chapter 5: Marks and Channels, 2015
M. Dörk, S. Carpendale and C. Williamson. VisGets: Coordinated Visualisations for Web-based Information Exploration and Discovery. In IEEE TVCG, 2008

Appendix 1: Suggested Theme: Child Poverty and Deprivation

Women and children in a rural community, illustrating the lived realities of child poverty. Photo source: https://data.unicef.org/

Why focus on child poverty?

Child poverty is multidimensional: it can involve low household income, material deprivation, limited access to services, and barriers to education, health, nutrition, water and sanitation, and protection.

Understanding how these dimensions vary across countries and regions can help reveal patterns of disadvantage and inform where interventions (including social protection) may have the greatest impact.

- For this year's practical, we encourage you to explore child poverty using UNICEF's State of the World's Children (SOWC) 2025 statistical tables, which provide comparable country-level indicators across multiple domains.
- The SOWC 2025 tables are especially suitable for asking open-ended questions.
- Exploring the child poverty theme through data
    - As part of this project, you should formulate your own open-ended question based on the dataset you choose to work with. If you use SOWC 2025, consider focusing on one domain (e.g., social protection and material deprivation) or combining a small number of indicators across tables (joined by 'Countries and areas') to explore relationships, outliers, and regional patterns.
    - When interpreting results, make sure to consider missing values, different reference years/ranges, and the limitations of cross-country comparisons. Your webpage and report should clearly state definitions/units and the years covered by the indicators you use.
    - We hope this theme sparks meaningful and insightful visualisations, encouraging you to develop your technical and analytical skills while exploring an urgent social issue.